

A cost of capital approach to determining the LGD discount rate

By J Larney, A Botha, GL Grobler and H Raubenheimer

Submission date 29 July 2025

Acceptance date 4 October 2025

ABSTRACT

Loss Given Default (LGD) is a key risk parameter in determining a bank's regulatory capital. During LGD estimation, realised recovery cash flows are to be discounted at an appropriate rate. Regulatory guidance mandates that this rate should allow for the time value of money, as well as include a risk premium that reflects the "undiversifiable risk" within these recoveries. Having extensively reviewed earlier methods of determining this rate, we propose a new approach that is inspired by the cost of capital approach from the Solvency II regulatory regime. Our method involves estimating a market-consistent price (that reflects market information) for a portfolio of defaulted loans, from which an associated discount rate may be inferred. We apply this method to mortgage and personal loans data from a large South African bank. The results reveal the main drivers of the discount rate to be the mean and variance of these recoveries, as well as the bank's cost of capital in excess of the risk-free rate. Our method therefore produces a discount rate that reflects both the undiversifiable risk of recovery recoveries and the time value of money, thereby satisfying regulatory requirements. This work can subsequently enhance the LGD component within the modelling of both regulatory and economic capital.

KEYWORDS

Loss given default, discount rate, risk premium, cost of capital, solvency

CONTACT DETAILS

Janette Larney, Centre for Business Mathematics and Informatics, North-West University, Private Bag X6001, Potchefstroom, 2520; National Institute for Theoretical and Computational Sciences (NITheCS), Stellenbosch, 7600; Email: janette.larney@nwu.ac.za

Arno Botha, Centre for Business Mathematics and Informatics, North-West University, Private Bag X6001, Potchefstroom, 2520; NITheCS, Stellenbosch, 7600

Gerrit Lodewicus Grobler, Department of Statistics, North-West University, Private Bag X6001, Potchefstroom, 2520

Helgard Raubenheimer, Centre for Business Mathematics and Informatics, North-West University, Private Bag X6001, Potchefstroom, 2520; NITheCS, Stellenbosch, 7600

1. INTRODUCTION

1.1 One of the key risk parameters used in determining a bank's regulatory capital under the Basel II Capital Accord from the Basel Committee on Banking Supervision (BCBS) (2006) is the loss given default (LGD). Denoted as the random variable $L \in [0,1]$, a loan's LGD is the fraction of its balance B_{τ_d} (or the exposure at default (EAD)) that was eventually lost on a defaulted account, as at the time of default τ_d . The LGD is often described as the "economic loss" on a financial instrument, which includes both material discount effects and material direct/indirect collection costs; see BCBS (2006: §460), itself corroborated by the European Parliament (2006). In nursing the strained relationship between the bank and the defaulted borrower back to health, collection staff can pursue a variety of remedial actions during this so-called *resolution/workout period*. From Van Gestel & Baesens (2009: 26–8), Finlay (2010: 11–3), and Botha (2021: §2.2), these remedial actions commonly include:

1. temporarily lowering instalments and/or zeroing interest rates;
2. extending a payment "holiday" during which instalments are suspended; and
3. restructuring the loan in the borrower's favour.

Successful collection efforts imply the distressed borrower repaying a newly-negotiated stream of cash flows over time and/or lump sums, both of which are called *receipts* in this work. These receipts should diminish the arrears balance and decrease the delinquency, ultimately "curing" the loan back to health; resulting in a zero credit loss $L=0\%$. Conversely, collection efforts may fail and further erode the credit relationship beyond repair. The bank will now initiate legal proceedings and/or foreclose on any available collateral, in recovering as much of the defaulted debt in record time. Any last receipts resulting from debt recovery are offset against the loan's last-known balance, with the non-zero remainder being written (or charged) off as a credit loss, i.e., $L>0\%$. Accordingly, a defaulted loan will resolve into either a *cured* or a *written-off* outcome, with the remaining unresolved cases being right-censored (pending an outcome). These ideas are illustrated in Figure 1.

1.2 Under the advanced internal rating based (IRB) approach of Basel, a bank can produce its own LGD estimates. From Schuermann (2004), Van Gestel & Baesens (2009: 217–22), and Baesens et al. (2016, §10), there are four common methods for estimating the LGD of either retail or corporate exposures. Of these methods, we shall focus on the more popular *workout* method, which caters to both retail and corporate exposures. At each month-end time period t , let X_{it} denote the net cash flow (or receipt minus all costs and benefits) of

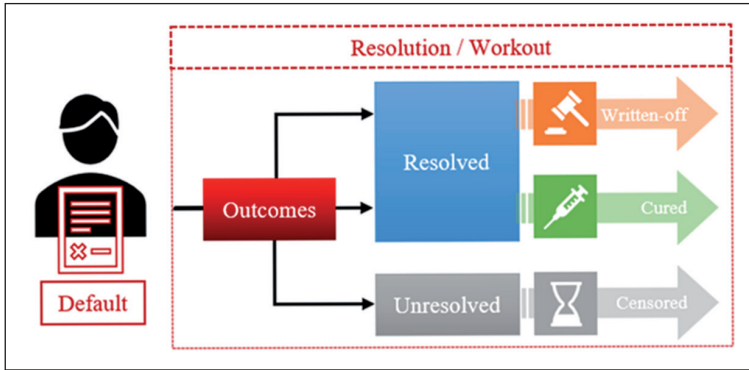


FIGURE 1. Illustrating the typical process of resolving a defaulted loan into either a written-off or cured outcome during the workout period. The remaining defaults are considered as right-censored. Source: Botha (2021: 59)

account $i = 1, 2, \dots, n$ over applicable time periods $t = \tau_d(i), \dots, \tau_r(i)$ for loan i , as observed from data. Similarly, let B_{it} record the month-end outstanding balance or utilised amount of a credit limit that was observed from data. For a written-off loan i , the realised loss $l_i(u)$ is then calculated at (and from) a given time u as the proportion of B_{iu} that was ultimately lost. This given time u can range from the moment of default $\tau_d(i)$ of loan i , up until its resolution/write-off time $\tau_r(i) \geq \tau_d(i)$; see Gürtler & Hibbeln (2013), Baesens et al. (2016, §10), and Scheule & Jortzik (2020). Using an appropriate factor v_m that discounts a quantity m periods backwards, the realised loss (given write-off) is expressed more formally as

$$l_i(u) = 1 - \frac{1}{B_{iu}} \sum_{t=u}^{\tau_r(i)} X_{it} v_{t-u} \quad \text{for } u = \tau_d(i), \dots, \tau_r(i) \text{ and } B_{ij} > 0. \quad (1)$$

1.3 In preparing the modelling dataset, Equation 1 is typically evaluated only once at the default time $\tau_d(i)$ of each resolved defaulted loan i that was eventually written off. By implication, these $l_i(\tau_d)$ values are considered as loan-level realisations from L across all written-off loans i . Finally, we illustrate the workout method in Figure 2 for a single hypothetical loan that was written off.

1.4 In Equation 1, selecting the appropriate discount rate in v_m can be non-trivial and even contentious in correctly accounting for both the time value of money and any inherent risks. Moreover, the cumulative effect of any discount rate on $l_i(\tau_d)$ becomes particularly pronounced for longer resolution periods, e.g., as in residential mortgages. Van Gestel & Baesens (2009: 220–1) noted that the most appropriate rate ought to be the “risk-free” rate r_f . Furthermore, and from a regulatory perspective, the BCBS (2005) offers principled (but limited) guidance on selecting an appropriate discount rate, which is arguably inspired by the seminal work of Markowitz (1952). In short, discount rates must reflect the time value of money (r_f) as well as include an appropriate risk premium δ (for “undiversifiable risk”),

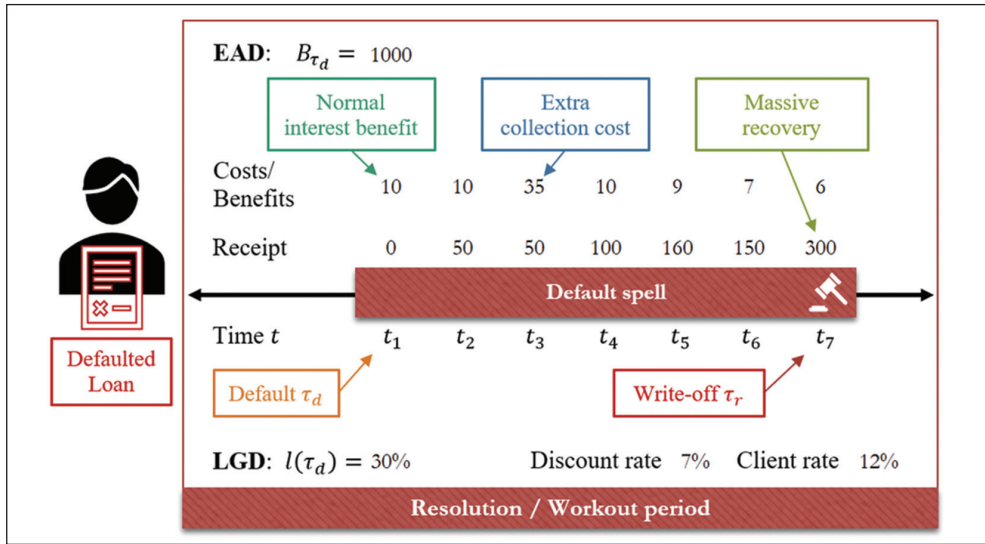


FIGURE 2. An example of using the workout method to calculate a written-off loan's LGD value from a bank's own credit data. Given a default balance of $B_{\tau_d} = 1000$ and net cash flows (receipts minus costs/benefits) over subsequent times $t = \tau_d, \dots, \tau_r$, the associated LGD is calculated as $l(\tau_d)$ at default time τ_d using Equation 1.

which compensates for the uncertainty of receiving X_{it} . We therefore structure this so-called Basel principle as follows, whereupon we rewrite the discount factor v_m in Equation 1 using the discount rate $r_d \in [0, 1]$; i.e.,

$$\text{Basel principle: } r_d = r_f + \delta \Rightarrow v_m = \frac{1}{(1 + r_d)^m}. \quad (2)$$

1.5 Some regulators have adopted this principle verbatim in selecting a discount rate, e.g., see the retail credit template of the South African regulator, the South African Reserve Bank (SARB) (2018: 20). Others have digressed somewhat, e.g., the European Union (EU) regulator, the European Banking Authority (EBA), proposed using a discount rate comprised of the primary interbank rate plus 5%; see EBA (2017). Though simple, this proposal deliberately ignores a bank's funding costs when assessing the uncertainty of receiving X_{it} ; itself affected by competent collection staff and associated indirect costs. Similarly, the UK regulator, the Prudential Regulation Authority (PRA), suggests a discount rate that consists of the Sterling Overnight Index Average (SONIA) as the interbank rate, plus 5%; see PRA (2019) and PRA (2020). In addition, the PRA imposes a minimum of 9% for discount rates used when estimating downturn LGD values. It is quite clear to us that there exists little consensus regarding even the prescription of r_d amongst regulators, and perhaps even less amongst practitioners.

1.6 As an alternative, we propose a cost of capital (CoC) approach for inferring the risk premium δ based on a market-consistent (or fair) price for defaulted exposures. The phrase “market-consistent price” is used in the general economic sense in that the price reflects overall market information. As inspired by the Solvency II (SII) regulatory regime for insurers (European Parliament, 2009), our CoC approach can produce a δ value that reflects the undiversifiable risk within X_{it} . Our work is closest to that of Altman & Pompeii (2003), Brady et al. (2006), Witzany (2009), and Jacobs (2012), wherein the discount rate was similarly inferred from market prices of defaulted debt, which typically includes corporate bonds and traded bank loans. These studies rely on the “informational efficient market” proposition of Fama (1970), which assumes that market prices incorporate all available information about future values. As such, the inferred discount rate should objectively reflect the market’s confidence on receiving X_{it} in future, provided that the market is sufficiently deep and liquid; see Jankowitsch et al. (2014). However, these procedures fail in the absence of such market prices, which is often the case for retail loan portfolios. Our CoC approach can generate such market-consistent (but synthetic) prices, driven by the cost of holding capital against uncertain cash flows.

1.7 The remainder of this paper is organised as follows. In Section 2, we provide an overview of the most relevant methods from literature in determining the LGD discount rate. We outline in Section 3 the fundamental ideas of the SII regulatory regime in calculating the cost of capital. Thereafter, the formulation of our CoC approach is provided in Section 4 towards calculating a market-consistent price (MCP) for a stream of uncertain cash flows X_{it} . In Section 5, we calibrate this CoC approach using two different credit datasets from a large South African bank: a secured mortgage portfolio and an unsecured personal loans portfolio; both of which are described. The CoC approach is illustrated in Section 6, thereby obtaining an MCP from which δ can be inferred, whereafter we conclude the study in Section 7.

2. EARLIER METHODS FOR DETERMINING THE LGD DISCOUNT RATE

2.1 In determining an appropriate v_m value for Equation 1, a few approaches exist in literature. Scheule & Jortzik (2020) studied and compared a few different methods of calculating a discount rate for LGD modelling, having focused on the robustness and empirical implications of these methods. These methods respectively use:

1. the loan’s contract rate at origination;
2. the loan’s expected return (ER);
3. the market’s average return on defaulted debt (RODD);
4. the bank’s cost of equity (or ROE);
5. the market equilibrium (ME) variant of ROE using Basel asset correlations; and
6. the weighted average cost of capital (WACC).

As the most popular, the first method simply uses the contract rate r_i of loan i when discounting the cash flows $X_i = \{X_{it_1}, X_{it_2}, \dots\}$ in Equation 1. This method was first proposed by Asarnow & Edwards (1995) in measuring the loss on defaulted corporate bonds held by

Citibank. In its simplest form, the contract rate $r_i \in [0,1]$ is constructed as the sum of a base rate ρ_c , a pre-selected profit margin m , and a factor f_g for loans $i \in S_g$ within each subset S_g of the portfolio that coincides with the g^{th} credit risk grade; i.e., the nominal annual rate $r_i = \rho_c + m + f_g$. A loan with greater credit risk therefore warrants a greater risk factor f_g , thereby increasing its “price” r_i ; see Thomas (2009, §3) and Phillips (2013) regarding risk-based pricing.

2.2 Given its availability at origination, the contract rate r_i is a natural and rather popular choice for discounting cash flows X_i when calculating realised LGD values. In fact, Scheule & Jortzik (2020) argued that its popularity is further entrenched by the two dominant accounting standards that guide impairment modelling: International Financial Reporting Standards (IFRS) 9 from the International Accounting Standards Board (IASB) (2014) and the current expected credit loss (CECL) framework from the Financial Accounting Standards Board (FASB) (2016). Both IFRS 9 and CECL require the use of the effective interest rate (EIR), which is the rate that exactly discounts an impaired loan’s future cash flows throughout its remaining expected life to the net carrying amount, i.e., the internal rate of return; see §5.4 and §B.5.4 of IFRS 9. Since future cash flows incorporate the r_i -based instalment, Scheule & Jortzik (2020) noted that the EIR tends to be similar to r_i . However, and despite its popularity, the authors note at least one major drawback of using r_i in discounting X_i . Since r_i embeds a loan’s expected loss via f_g , the presumption is that the default event is uncertain (and hence unrealised) when pricing said loan at origination. However, calculating the realised LGD of loan i is predicated upon first realising the default event, whereupon the subsequent cash flows X_i are observed. This implies that using r_i in discounting X_i will overcompensate for the now-realised default risk, which can lead to artificially greater LGD values. Doing so disagrees with the Basel principle in that, as explained in both Maclachlan (2005) and Jacobs (2012), the discount rate should only compensate for the uncertainty of receiving X_i post-default, i.e., the so-called *resolution risk*. Put differently, the contract rate exceeds the expected return when conditioning for default, as demonstrated by Scheule & Jortzik (2020) in their second ER method.

2.3 Another simple approach for informing r_d is the third so-called RODD method, which is based on the price movements of tradable junk bonds. Given the price $\mathcal{B}_{it}$ of such a defaulted bond i at time t , the realised return ρ_i is expressed over the associated resolution period $\tau_r - \tau_d$ as

$$\rho_i = \left(\frac{\mathcal{B}_{i\tau_r}}{\mathcal{B}_{i\tau_d}} \right)^{\frac{1}{\tau_r - \tau_d}} - 1. \quad (3)$$

2.4 The exact moment of default may not be known for a bond, though Scheule & Jortzik (2020) remark that the bond price between 30 or 45 days past default is typically taken as a proxy for $\mathcal{B}_{i\tau_d}$. Having calculated the returns ρ_i for a set of defaulted bonds S_b , the expected return $\mathbb{E}(\rho_r)$ is estimated using the sample mean, which is then used as the value for the discount rate r_d in v_m within Equation 1. The RODD method is therefore summarised as

$$\text{RODD method: } \mathbb{E}(\rho_r) \approx \frac{1}{|\mathcal{S}_b|} \sum_{i \in \mathcal{S}_b} \rho_i \quad \therefore r_d = \mathbb{E}(\rho_r) \quad (4)$$

2.5 While certainly simplistic, the RODD method can yield a negative value for r_d , which would be nonsensical in discounting the cash flows X in Equation 1. This can happen when the prices of most bonds within $\mathcal{S}_b$ have decreased at their respective resolution times τ_r , i.e., $\mathcal{B}_{i\tau_r} < \mathcal{B}_{i\tau_d}$ holds for most i .

2.6 Alternatively, the discount rate r_d can be based more directly on the bank's funding costs by using the fourth ROE method from Scheule & Jortzik (2020). Let ρ_e denote the returns over time for a hypothetical instrument that is comparable to a portfolio of defaulted loans, subject to resolution risk. Then, the expected return on equity $\mathbb{E}(\rho_e)$ is estimated using the classical capital asset pricing model (CAPM) from Sharpe (1964), which is formalised as

$$\text{ROE method (CAPM): } \mathbb{E}(\rho_e) = r_f + \beta \cdot \underbrace{\mathbb{E}(r_m - r_f)}_{\text{Market risk premium (MRP)}} \quad \therefore r_d = \mathbb{E}(\rho_e), \quad (5)$$

where r_f is the (known) risk-free rate and r_m is the (unknown) market-related return of a similar portfolio. In setting either parameter, Scheule & Jortzik (2020) explained that r_m is often replaced with the equity returns on the local share market, e.g., the All-Share Index (ASI) from the Johannesburg Stock Exchange (JSE) (2023) that tracks about 99% of eligible securities.¹ In fact, Dimson et al. (2011) examined such equity returns relative to those of either government bonds or treasury bills (both of which can substitute r_f) and found that these “excess returns”, or estimates of $\mathbb{E}(r_m - r_f)$, broadly averaged 3–8% using 110 years of data across 19 countries.

2.7 The β within the ROE method from Equation 5 measures the sensitivity of the instrument's returns to the variation of excess market returns. From Fama & French (2004) and Scheule & Jortzik (2020), this so-called *market beta* is expressed as

$$\beta = \frac{\kappa(r_m, \rho_e) \sigma_{\rho_e}}{\sigma_{r_m}}, \quad (6)$$

where $\kappa(X, Y)$ is the Pearson correlation between two random variables X and Y , and σ_X is the standard deviation of X . The authors noted previous studies that have used the New York University (NYU) Bond Index and 90 defaulted United States (US) bonds as two candidates for ρ_e , whereupon they obtained β values between 35–40%.

2.8 Maclachlan (2005) and Scheule & Jortzik (2020) provided another interpretation of the correlation $\kappa(r_m, \rho_e)$ in Equation 6 that leverages a method by which the regulatory

1. JSE (2023). All Share Index. Johannesburg Stock Exchange. <https://www.jse.co.za/all-share-j203> (Retrieved 14 August 2023)

capital of a bank is normally calculated. They use the asymptotic single-risk factor (ASRF) model from Vasicek (2002) and extended by Gordy (2003); itself codified in the Basel framework from the BCBS (2019). In particular, the ASRF model supposes that a corporate exposure i defaults once its overall asset value Y_i falls below a certain threshold. This Y_i value is then structurally decomposed into:

1. a portfolio-level *systematic* or market factor V_m that affects all exposures equally; and
2. an *idiosyncratic* risk factor Z_i specific to i .

For the loan pair i and j , the associated random variables (Y_i, Y_j) are assumed to be bivariate normally distributed with standard normal marginals and with equal pairwise correlations $\kappa(Y_i, Y_j) = \kappa$, thereby accounting for correlated defaults when asset values move together. Finally, Y_i is expressed using κ with a Gaussian copula as $Y_i = \sqrt{\kappa} \cdot V_m + \sqrt{1 - \kappa} \cdot Z_i$; see Botha (2021, §2.5). The key economic insight from Maclachlan (2005) is that $\sqrt{\kappa}$ behaves like $\kappa(r_m, \rho_e)$ when substituting ρ_e with the market value of a defaulted exposure i . Accordingly, the authors provided a variant of the ROE method (or the fifth ME method) by redefining the β within the CAPM from Equation 5 as a *defaulted debt beta*, expressed for firm i as

$$\text{ME variant of ROE: } \beta_i = \frac{\sqrt{\kappa} \cdot \sigma_i}{\sigma_{r_m}}, \quad (7)$$

where σ_i denotes the standard deviation of the i^{th} firm's asset value over time. The standard formulae for κ can be used as prescribed by the Basel framework in BCBS (2019, §CRE31), e.g., a fixed 15% for retail residential mortgages, or an exponentially-weighted interpolation between 12–24% for corporate exposures based on their default risk. Accordingly, Maclachlan (2005) illustrated this equilibrium-method with $\beta_i \in [0.4, 1]$ across a range of regulatory asset classes, whereupon he obtained r_d values of between 2.4–6%.

2.9 On its own, the CAPM has a few unrealistic assumptions that undermine its tractability in practice, which compromises the ROE method. Fama & French (2004) examined some of these assumptions empirically and corroborated previous findings that ultimately rejected the CAPM. In particular, the CAPM assumes complete agreement amongst investors on the joint distribution of asset returns and its constituents, which assumes that $\mathbb{E}(\rho_e)$ from Equation 5 is indeed linearly related to both the risk-free rate r_f and the market beta β . The authors tested this assumption by regressing a time series of monthly portfolio returns on estimates of β , as calculated using Equation 6. By implication, the intercept of such a simple regression model should approach r_f , while the coefficient on β should approximate the excess market return $\mathbb{E}(r_m - r_f)$. While a positive linear relationship does exist between average returns and market betas, the real-world relationship is significantly “flatter” than predicted by the CAPM. In fact, these studies consistently find that the intercept is greater than the average r_f (as proxied by the return on a 1-month US Treasury Bill), while the coefficient on β is less than the average excess market return (as proxied by the return on common US stocks minus that of a 1-month Treasury Bill). Fama & French (2004) also reviewed other studies in

showing that β is not the only factor that explains expected returns. Ratio-type variables like earnings price, debt equity (or leverage), and book-to-market can challenge the sufficiency of β within the CAPM in predicting expected returns. This result is perhaps unsurprising since these variables embed information about the market's expectation of future returns, simply by incorporating the share price. As a viable alternative to the CAPM, the authors accordingly propose their *3-factor Fama–French model*, which includes both firm size and book-to-market value as additional factors.

2.10 While the ROE method and its ME variant may be alluringly simple, Scheule & Jortzik (2020) noted that funding costs alone do not reflect the risk profile of subsequent lending activities, especially not the resolution risk thereof (cures/write-offs) given default. Within the sixth so-called WACC method, the authors combine both the regulatory capital of a bank and its funding costs into a single rate, given that regulatory capital is a reasonable measure of systematic risk. In particular, let ρ_c represent the return of an instrument that is comparable to a defaulted exposure subject to resolution risk. This return ρ_c is itself a blend of the costs of equity and debt-based funding, denoted respectively as ρ_e and ρ_d . The CAPM from Equation 5 can be used to estimate the expected equity return $\mathbb{E}(\rho_e)$, while $\mathbb{E}(\rho_d)$ can be derived from the weighted average cost of bank liabilities. The novelty lies in the so-called “funding weights” by which the estimates for $\mathbb{E}(\rho_e)$ and $\mathbb{E}(\rho_d)$ are eventually blended into the estimate for $\mathbb{E}(\rho_c)$. Given default, the equity-based capital ratio e_{τ_d} may be loosely defined as the amount of capital held for the unexpected loss, divided by the post-default loan value (or expected recovery); i.e.,

$$e_{\tau_d} = \frac{[\mathbb{E}(L | \text{Downturn}) - \mathbb{E}(L)] \cdot \text{EAD}}{[1 - \mathbb{E}(L)] \cdot \text{EAD}} = \frac{\text{Bank Capital}}{\text{Remaining Loan Value}}, \quad (8)$$

where $\mathbb{E}(L | \text{Downturn})$ is the expected LGD during an economic downturn. Finally, the loan-specific e_{τ_d} estimate is used as the weight for blending the bank-level funding costs, summarised as

$$\text{WACC method: } \mathbb{E}(\rho_c) = e_{\tau_d} \mathbb{E}(\rho_e) + (1 - e_{\tau_d}) \mathbb{E}(\rho_d), \quad (9)$$

whereupon r_d is set to the estimate of $\mathbb{E}(\rho_c)$ and used in v_m within Equation 1. Naturally, the ROE method is simply a special case of the WACC method by setting the capital ratio $e_{\tau_d} = 100\%$ in Equation 9, whereupon $\mathbb{E}(\rho_c) = \mathbb{E}(\rho_e)$.

2.11 Scheule & Jortzik (2020) compared the aforementioned methods both qualitatively and quantitatively using global credit data across different geographies. The ROE method from Equation 5 often yielded the highest mean discount rate r_d of 7.07% with a resulting mean realised LGD $l(\tau_d)$ of 24.95%. The contract rate method gave a significantly lower mean r_d of 4.91%, coupled with a mean $l(\tau_d)$ of 23.82%. The ME variant from Equation 7

tendered a lower mean r_d of 4.13% that yielded a mean $l(\tau_d)$ of 23.35%. As a variant of the contract rate method, the ER method produced a lower mean r_d of 3.72%, which resulted in a mean $l_i(\tau_d)$ of 23.16%. The WACC method yielded an even lower mean r_d of 3.26%, which produced a mean $l(\tau_d)$ of 22.89%. Lastly, and as a baseline, simply using the risk-free rate r_f at default time τ_d gave the lowest mean r_d of 2.36% and a resulting mean $l(\tau_d)$ of 22.41%. The last method's low rank might very well suggest that the resulting mean LGD might be underestimated.

2.12 Another notable method for estimating r_d is the market-sensitive *cost of risk capital* approach from Witzany (2009), which might reflect the uncertainty of receiving cash flows. In particular, the author developed an iterative approach that is based on the CAPM as departure point, whilst blending in ideas from the ASRF model under the Basel framework. This approach presupposes that realised loss rates can themselves be decomposed using a Gaussian copula between a systematic component and an idiosyncratic component, both of which are assumed to be independent standard normally distributed variables. Having calculated the expectation hereof using numerical procedures, the author provided an expression for the unexpected loss based on the Value-at-Risk (VaR) at some probability level for market risk. This quantity then informs the LGD risk capital, which is multiplied by the market-implied cost of holding such capital towards obtaining an associated r_d . In implementing this approach, a starting guess for r_d is needed, whereupon the realised loss rates are obtained and the rest of the method is calibrated accordingly, which ultimately produces a new r_d value. This iterative procedure continues until the difference between subsequent r_d values becomes suitably small. Our method shall borrow elements hereof in determining r_d using the market-sensitive cost of capital.

2.13 In determining a suitable risk premium δ , only a few other studies have investigated the relation between the market risk premium (MRP) under the CAPM from Equation 5 and the “cost of risk”, or *economic capital*. Maclachlan (2005), who developed the aforementioned ME variant of the ROE method, described economic capital as a risk charge held for the difference between a portfolio's maximum diminution in market value and its expected value. Even for defaulted exposures, it is deemed appropriate to hold economic capital in covering the systematic uncertainty within recovery cash flows; see Frye et al. (2000), BCBS (2003), Düllmann & Gehde-Trapp (2004) and Caselli et al. (2008). In extending this ME method from Maclachlan (2005), the authors Gibilaro & Mattarocci (2007) tweaked its reliance on the single risk factor from the ASRF model, and replaced it with a multi-factor model that can better explain the variance in the LGD. But to the best of our knowledge, a closed-form expression for economic capital is not widely known, except for the more esoteric proposals from Tasche (2004) and Weißbach & von Lieres und Wilkau (2010) using numerical procedures. In particular, Tasche (2004) proposed using a variant of the ASRF model wherein the LGD is treated as a random variable towards incorporating probability of default (PD)-LGD correlation. As mentioned earlier, in the ASRF model, defaults are driven by a random variable Y , called a risk-factor, that represents the overall

asset value. In the Tasche model, Y not only drives the default event, but also the size of the loss incurred. LGD is then expressed as $L = \mathbb{I}(Y \leq \xi)G(Y)$, where ξ is a chosen threshold value and G is an increasing or decreasing function. Expressing LGD as a function of the default event in this way was first proposed in a two-factor model by Pykhtin (2003), where the default event and the loss were driven by two random variables. However, the Tasche model does not require estimating additional parameters and fits rather comfortably within the mathematical machinery of the existing Basel framework. Given its simplicity and ease of implementation, we shall use the Tasche model in estimating economic capital as an input within our CoC approach.

3. THE COST OF CAPITAL UNDER THE SOLVENCY II REGIME

3.1 Solvency II is a regulatory regime aimed at insurance and reinsurance undertakings in the European Union (EU), as enacted by the European Parliament (2009). Several non-EU countries, including South Africa,² Australia and Singapore have adopted SII-equivalent risk-based capital regimes, as discussed by Ferguson (2009). Similar to the Basel framework for banks, SII is based on the principle that institutions should hold enough capital to cover the potential (unexpected) future losses that could result from their risk-inducing activities. However, SII and Basel differ in setting the required capital level, as well as in the specific rules and regulations that they impose.

3.2 One of the key components of SII is the so-called “economic balance sheet” approach, where capital requirements are based on a balance sheet that reflects the economic value of a firm’s assets and liabilities. Both assets and liabilities must be valued at *fair value*, i.e., “the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date”, as defined by IASB (2011). In most cases, the fair value is simply the current market value of an asset or liability, or alternatively, the market value of a replicating portfolio (Boekel et al., 2009). If such a market value does not exist (i.e., the specific market is not deep, liquid, or transparent), then SII requires valuation to be at least market-consistent. To achieve such consistency, the “best estimate plus risk margin” approach has been adopted, where the best estimate of the present value (or BEPV) of future cash flows is adjusted by a risk margin (RM), or the market value margin. In turn, this particular BEPV is defined by SII as the “probability-weighted average of future cash flows”, which incorporates the time value of money via discounting at least at the risk-free rate.

3.3 The idea with the RM is that a potential buyer of liabilities should be compensated for the cost of holding risk capital (economic or regulatory) in excess of the BEPV of the associated cash flows. Doing so will protect this buyer against the prospect of the BEPV

2. Solvency Assessment and Management (SAM) represents a “Solvency II equivalent” risk-based supervisory approach for the prudential regulation of South African long-term and short-term insurers. See <https://tinyurl.com/tawdn8mz>

value being too low. Similarly, an asset buyer should be compensated for the uncertainty surrounding the BEPV of the asset's future cash inflows. This RM therefore compensates the providers of capital, whose funds are used to absorb the risk associated with uncertain cash flows. The RM is then subtracted from the BEPV of assets (or added to the BEPV of liabilities) to determine the MCP of the underlying cash flows. Regarding its calculation, the RM is expressed as the cost of holding the required amount of capital over the run-off period of the risky cash flows. The base principle is that the price of an asset or liability should reflect the cost of holding risk capital against inherent risks over its lifetime. This is the so-called CoC approach from the SII regime, which is further discussed by Pelkiewicz et al. (2020).

3.4 In determining an MCP for a portfolio of defaulted debt, one needs to calculate the price that a potential investor would be willing to pay for the risky cash flows, or recoveries. As per the CoC approach, these cash flows that represent the recoveries expected from the defaulted loans are first discounted at the risk-free rate r_f . From the resulting BEPV, the cost of holding risk capital (against the risk of recovering less than the expected amount) is then deducted. Any such capital required at any future time t is also discounted at the r_f back to the transaction date. Doing so reflects the fact that the capital buffer will be invested in assets that earn at least at r_f . In calculating the RM, the CoC approach then requires an estimate of the required capital level as well as its cost (in the form of a rate). A few methods exist for determining this capital level, e.g., the Tasche (2004) model for determining economic capital. More importantly, and regarding the capital cost rate, consider the net cost of holding risk capital, which is the marginal cost of raising (and holding) risk capital, minus the interest that can be earned on such capital at r_f . In this study, we consider a bank's own cost of equity in excess of r_f as a capital cost rate, which represents a realistic view of the associated cost of raising and holding the necessary risk capital. Such a rate could be prescribed by the regulator, in a similar way that SII prescribes an CoC rate of 6% for insurance and reinsurance undertakings.

4. TOWARDS A MARKET-CONSISTENT RISK PREMIUM

In Section 4.1, we provide a numeric algorithm for finding a plausible δ value based on the MCP. This algorithm accepts estimates of the overall cost of capital over time, for which we provide an illustration in Section 4.2 using the work of Tasche (2004).

4.1 Determining the risk premium given economic capital estimates: a generic algorithm

4.1.1 Our aim is to determine the MCP for a portfolio of defaulted debt, and from this, calculate the associated risk premium δ for use within Equation 2. This process requires an appropriate amount of economic capital C_t over discrete time $t = 1, 2, \dots$, as discussed in Section 2. However, the economic capital (EC) is generally a function of the workout LGD, which in turn relies on the discount rate $r_d = r_f + \delta$; i.e., C_t is at least a function of δ . Accordingly, and using a given capital vector $C = C_t(\delta)$ over t , we define the risk margin R for a portfolio of defaulted loans as the function

$$R(C, \delta) = \sum_{t=1}^{\tau_A} \frac{c C_t(\delta)}{(1+r_f)^t} \quad (10)$$

where $c \in [0,1]$ is an appropriate cost of capital rate, and τ_A is the maximum time spent in default across the portfolio, i.e., $\tau_A = \max_{1 \leq i \leq n} \{\tau_r(i) - \tau_d(i)\}$. Let Y denote the present value of the portfolio-level recoveries $X_t = \sum_{i=1}^n X_{it}$ for loans $i = 1, \dots, n$ at each period t , where $r_d = r_f + \delta$ is used as the discount factor. Since r_f is treated as a constant, we express the so-called *discounted recoveries* Y as the function

$$Y(\delta) = \sum_{t=1}^{\tau_A} \frac{X_t}{(1+r_f+\delta)^t}, \quad (11)$$

4.1.2 Equipped with these two components Equations 10 and 11, our objective is to find the value of $\delta \geq 0$ that minimises the squared difference between the MCP and the $Y(\delta)$. More formally, our objective function f is

$$f(\delta, R, Y) = \left(\underbrace{[Y(0) - R(C, \delta)]}_{\text{MCP}} - Y(\delta) \right)^2, \quad (12)$$

and we are interested in finding

$$\delta_* = \underset{\delta \geq 0}{\operatorname{argmin}} f(\delta, R, Y). \quad (13)$$

4.1.3 However, note that $C_t(\delta)$ is already a function of δ , which is now itself a function of $C_t(\delta)$ given Equation 13, a circular reference. We deliberately break this circularity by pre-calculating R using a preset (but amendable) value of δ towards calculating C within a broader iterative approach. Doing so allows the $Y(\delta)$ term within Equation 12 to be varied when finding δ in Equation 13, whereupon a new value for R can be calculated in restarting the process. Accordingly, we provide an iterative procedure in Algorithm 1 by which Equation 13 can be implemented in finding a δ value. This algorithm has the following general steps:

1. Set an initial value for $\delta = \delta_0$ and calculate the economic capital $C^* = C_t(\delta_0)$ across all t accordingly.
2. Calculate the risk margin $R^* = R(C, \delta_0)$ from Equation 10 using $C = C^*$ and $\delta = \delta_0$.
3. Calculate a baseline discounted recoveries $Y_0 = Y(\delta)$ from Equation 11 for $\delta = 0$.
4. Find δ_i by minimising f from Equation 12 but setting $R = R^*$, i.e.,

$$\delta_i = \underset{\delta \geq 0}{\operatorname{argmin}} \left\{ [Y_0 - R^*] - Y(\delta) \right\}^2. \quad (14)$$

5. Update $\delta = \delta_i$ and repeat steps 1 to 4 until δ_i converges, given a preset convergence criterion.

ALGORITHM 1. Pseudocode for finding a δ value via optimisation of Equation 13

Input: c, τ_A, r_f	▷ Cost of capital c ; maximum time spent in default τ_A ; risk-free rate r_f
Input: X_t	▷ Recovery cash flows X_t over discrete time $t = 1, \dots, \tau_A$
Input: $\tilde{C}(a, b)$	▷ Function for calculating EC vector $C = C_t(a)$ using discount rate a across $t = 1, \dots, b$
Output: δ^*	▷ Outputs an estimate δ^* that minimises Equation 13
1: $\epsilon \leftarrow 10^{-4}$	▷ Tolerance level for small values
2: $\phi^* \leftarrow 10^4$	▷ Initial error value
3: $\delta^* \leftarrow \delta_0$	▷ Initial value δ_0 towards finding δ^* iteratively
4: $Y_0 \leftarrow \tilde{Y}(\delta, \tau_A, r_f)$	▷ Calculate Equation 11 for $\delta=0$, implemented here as $\tilde{Y}$ with other inputs τ_A and r_f
5: while $\phi^* > \epsilon$ do	▷ Convergence criterion: loop until the error ϕ^* becomes suitably small
6: $C^* \leftarrow \tilde{C}(\delta^*, \tau_A)$	▷ Estimate EC vector for δ^* , implemented here as $\tilde{C}$ with other input τ_A
7: $R^* \leftarrow \tilde{R}(C^*, c, \tau_A, r_f)$	▷ Calculate Equation 10 for C^* , implemented here as $\tilde{R}$ with other inputs c, τ_A , and r_f
8: $f := \text{function}(\left[Y_0 - R^* \right] - \tilde{Y}(\delta^*, \tau_A, r_f))^2$	▷ Initiate f from Equation 12 using R^* and Y_0 as “static” inputs
9: $\delta_i \leftarrow \text{minimise}(f, \delta^* \geq 0)$	▷ Implement Equation 13, i.e., find $\delta^* \geq 0$ that minimises f
10: $\phi^* \leftarrow \delta^* - \delta_i $	▷ Update convergence criterion
11: $\delta^* \leftarrow \delta_i$	▷ Update δ^*
12: end while	

4.1.4 Having applied Algorithm 1 in calculating δ , it so happens that $r_f + \delta$ is the internal rate of return when equating MCP to the discounted cash flows X_t . This represents the fair return offered by investing in the portfolio of defaulted loans with δ serving as compensation for the uncertainty associated with the income stream. This δ quantity is itself driven by the cost of raising and holding the required EC amount in respect of the uncertain cash flows. As such, we argue that this risk premium δ is appropriate for inclusion in the LGD discount rate r_d from Equation 2.

4.2 Estimating economic capital using the single-factor Tasche model

4.2.1 According to the Basel framework, the EC amount for covering total loss L' is calculated as

$$\mathbb{E}[L' | V_m = \Phi^{-1}(\alpha)] - \mathbb{E}[L'], \quad (15)$$

where α is chosen at a suitably high level, and Φ^{-1} is the inverse cumulative distribution function (CDF) of a standard normal variate Z . Put differently, the EC amount is calculated as the difference between the VaR of the loss distribution and the expected loss of the loan

portfolio, as discussed by Botha (2021: 54–5). Then, and as outlined in Section 2, Tasche (2004) proposed a method that incorporates the PD-LGD dependency by expressing total loss L' as

$$L' = \mathbb{I}(W \geq \Phi^{-1}(1-p)) F_D^{-1} \left(\frac{\Phi(W) - 1 + p}{p} \right).$$

where $W = \sqrt{\kappa}V_m + \sqrt{1-\kappa}Z$ is the loss, κ is a correlation coefficient, F_D is the c.d.f. of the loss distribution, and p is the probability of default.

4.2.2 While the expectation $\mathbb{E}[L']$ in Equation 15 is simply calculated as $p\mathbb{E}[L' > 0]$, the conditional loss term is expressed as

$$\mathbb{E}[L' | V_m = \Phi^{-1}(\alpha)] = \int_a^\infty \varphi(z) F_D^{-1} \left(\frac{\Phi(\sqrt{\kappa}\Phi^{-1}(\alpha) + \sqrt{1-\kappa}z) - 1 + p}{p} \right) dz, \quad (16)$$

where

$$a = \frac{\Phi^{-1}(1-p) - \sqrt{\kappa}\Phi^{-1}(\alpha)}{\sqrt{1-\kappa}}$$

and φ is the density of a standard normal variate. By using Gauss quadrature numerical integration, Tasche (2004) proposes approximating the integral in Equation 16 with

$$\mathbb{E}[L' | V_m = \Phi^{-1}(\alpha)] \approx \frac{p}{2\sqrt{1-\kappa}} \sum_{i=1}^5 w_i H(t_i, \Phi^{-1}(\alpha)); \quad (17)$$

where

$$H(t, \Phi^{-1}(\alpha)) = \frac{\varphi \left(\frac{\Phi^{-1}(p(t+1)/2 + 1 - p) - \sqrt{\kappa}\Phi^{-1}(\alpha)}{\sqrt{1-\kappa}} \right)}{\varphi(\Phi^{-1}(p(t+1)/2 + 1 - p))} F_D^{-1} \left(\frac{t+1}{2} \right),$$

and the weights w_i and nodes t_i are given by

$t1 = -0.9061798459$	$w_1 = 0.2369268851$
$t2 = -0.5384693101$	$w_2 = 0.4786286705$
$t3 = 0$	$w_3 = 128 / 225$
$t4 = -t2$	$w_4 = 0.2369268851$
$t5 = -t1$	$w_5 = 0.2369268851$

4.2.3 In calculating the EC using the Tasche model, let the loss distribution F_D be a beta distribution with shape parameters a and b , which aligns with Tasche (2004). Despite its flexibility, some studies have demonstrated the beta distribution to be inappropriate since it cannot readily capture the bimodal shape of the LGD distribution; see Calabrese

& Zenga (2010), Huang & Liang Xie (2012) and Larney et al. (2022). However, despite its shortcomings, the beta distribution remains a popular and rather practical choice given its ability to assume many distributional shapes in modelling the LGD parametrically. It is also fairly straightforward to estimate its two parameters a and b by ways of moment matching, provided with the sample mean and sample variance of realised loss data. Finally, we set $C_t(\delta)$ equal to this EC amount, which is determined at the start of each of the run-off years for a loan portfolio at the 99.9% confidence interval. Doing so agrees with Basel requirements, as outlined by the BCBS (2004: 11–2]. While we have used the Tasche model in determining EC, a bank should consider its own strategic objectives and risk appetite when deciding the appropriate risk measure, confidence level, and time horizon to use.

5. A DESCRIPTION AND ANALYSIS OF REAL-WORLD LOSS DATA

5.1 We illustrate our CoC approach by applying it to two retail credit datasets provided by a large South African bank. The first dataset comprises recovery data of an unsecured personal loans (PL) portfolio, whereas the second dataset contains secured residential mortgage loans (ML). Each dataset includes cash flows (both positive and negative) received on accounts that defaulted during October 2005 to September 2011. This six-year period also includes an economic downturn period of two years, during which each portfolio suffered the highest default rates. In particular, the downturn periods respective to the PL and ML portfolios are December 2007 to November 2009 and September 2007 to August 2009, both of which coincide approximately with the global financial crisis of 2008; see Olbrys (2021). In estimating the risk-free rate r_f , we first interpolate between the two closest available tenors on the South African bond yield curve and then average the interpolated rates for all trading days within the relevant reference period. For example, the mean workout period for the full period on personal loans is 4.67 months. Therefore, for each day within our observed period, a risk-free rate is obtained by interpolating between the rates observed at a 3-month and 6-month tenor on the yield curve. The estimated risk-free rate of 6.37% is then obtained as the arithmetic average of these daily rates over the observed period.

TABLE 1. Summary statistics for PL and ML portfolios over the full and downturn periods

	n	$\bar{L}$	$\hat{\sigma}$	Max workout period	Mean workout period	r_f
Personal loans						
Full	121, 151	0.749	0.290	87 months	4.67 months	6.37%
Downturn	60, 032	0.760	0.273	80 months	4.57 months	8.93%
Mortgage loans						
Full	126, 580	0.256	0.366	113 months	25.64 months	6.88%
Downturn	60, 779	0.293	0.372	90 months	33.59 months	8.49%

5.2 The summary statistics for the two portfolios are shown in Table 1. Besides the difference in the average LGD ($\bar{L}$), note the significant difference in the averages of the

workout periods between the two portfolios. Given the lack of collateral, banks typically have stringent write-off criteria for unsecured PLs that shorten the workout process, which contrasts with that of asset-secured MLs. Consequently, the average LGD for the PL portfolio is also much higher than that for the ML portfolio. The latter's lower average LGD is however accompanied by a higher standard deviation ($\hat{\sigma}$). Interestingly, the means and standard deviations for the PL portfolio are very similar for the two periods under consideration, whereas the average LGD in respect of ML in the downturn period is somewhat higher at 29.3% compared to 25.6% over the full period. In Figures 3 and 4, we show the distribution of the aggregated LGD data over the full period, respectively for the PL and ML portfolios. As expected, both portfolios have a bimodal shape in the distribution of their LGD values, though the degree differs by portfolio type. In particular, the ML portfolio has a substantial proportion of cured accounts that result in zero losses, whereas these cases constitute only a minor proportion for the PL portfolio. Accordingly, the higher standard deviation of the ML portfolio is mainly attributed to the high proportion of these cured loans.

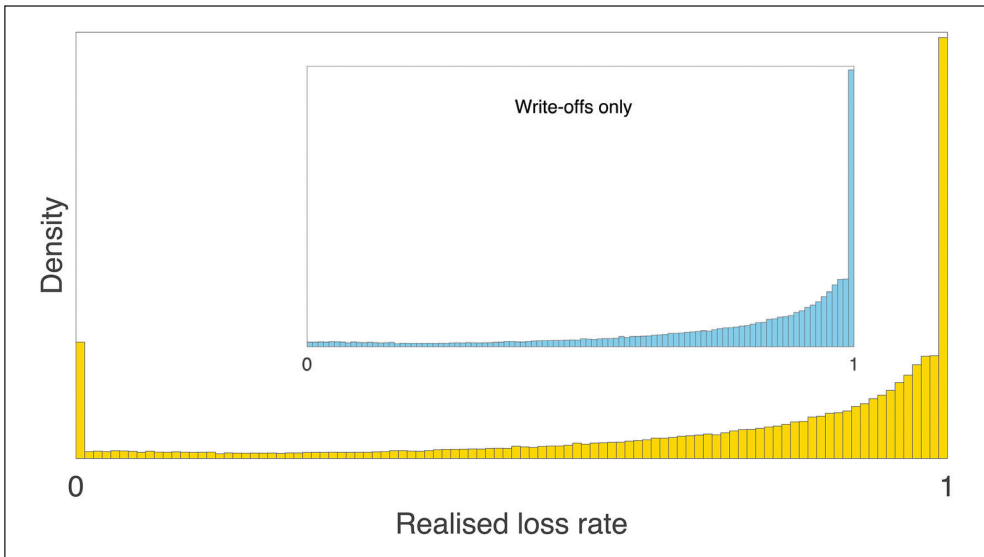


FIGURE 3. Histogram of realised loss rates for the unsecured PL portfolio over the full reference period. The outer graph shows all data (including 'cures'), whilst the inset graph depicts the distribution of those loans that were written off.

5.3 In Figure 5, we show the proportions of LGD observations over time at various points over the interval $[0,1]$. Evidently, the ML portfolio experienced a marked drop in the proportion of cured loans during the downturn period, and this proportion did not return to pre-downturn levels before the end of the reference period. A similar pattern emerges in the PL portfolio, where the proportion of total losses continue to increase well beyond the downturn period, after which it stabilises. This result suggests that the effects of the economic downturn continued well into 2011. The ML data suggests that the recovery of residential

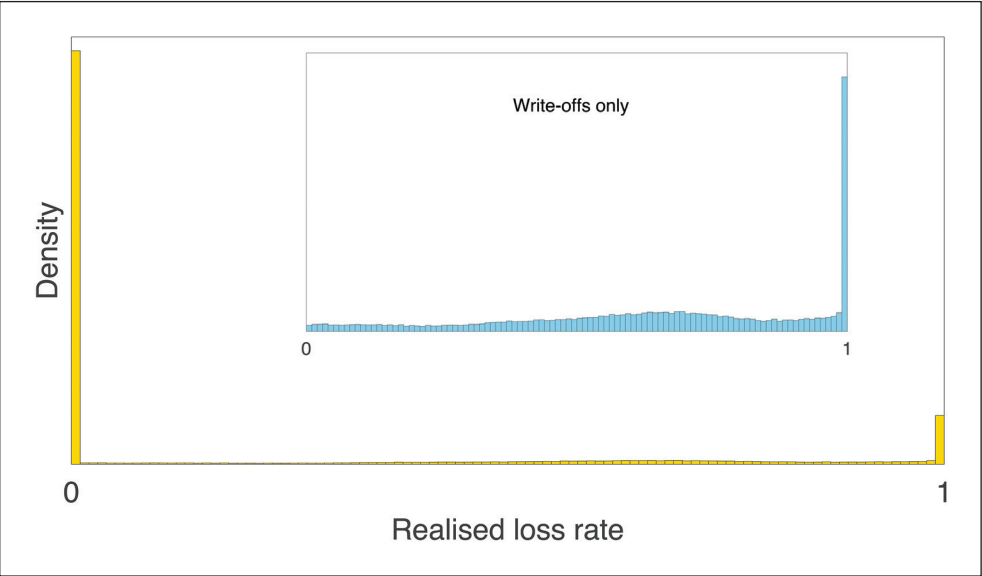


FIGURE 4. Histogram of realised loss rates for the secured ML portfolio over the full reference period. Graph design follows that of Figure 3.

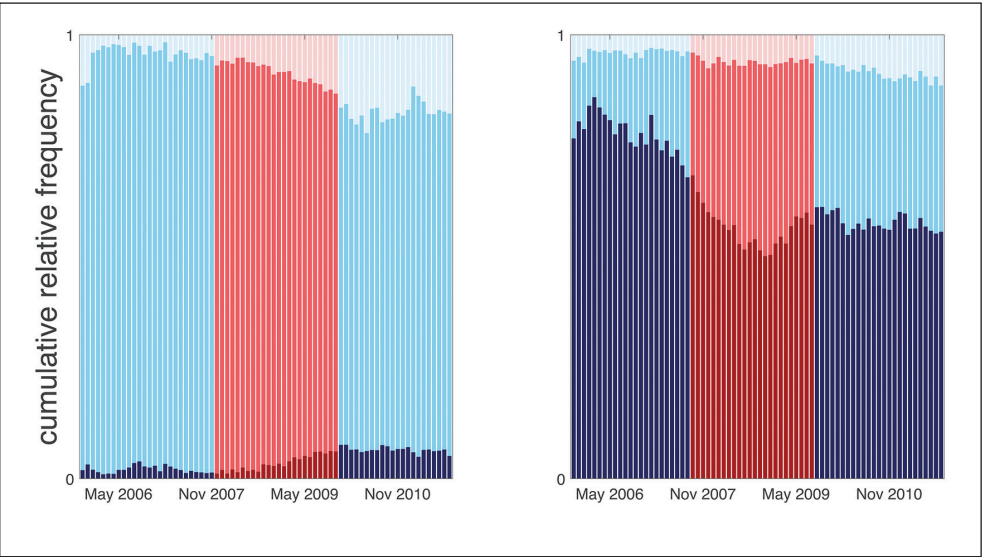


FIGURE 5. Proportion of LGD observations over time at 0 (dark blue), on the interval (0,1) (medium blue), and at 1 (light blue). Proportions are shown with respect to the PL portfolio (left panel) and the ML portfolio (right panel), whilst downturn months are shaded in red.

property prices (collateral) lagged behind economic recovery, resulting in greater loan-to-value ratios and fewer defaulted accounts with full recoveries. For both PLs and MLs, the trends could also indicate a change in the bank's underwriting standards. Following the severe losses suffered by banks during the 2008 global financial crisis, many banks adjusted their underwriting standards, as reported by the Federal Advisory Council and Board of Governors (2015) for the US market. This factor might also be at play in our dataset.

6. AN ILLUSTRATION OF DETERMINING THE RISK PREMIUM USING THE COST OF CAPITAL APPROACH

6.1 We first consider a cost of capital of $c=7\%$, as derived from the average cost of equity of approximately $14\%^3 \approx 7\% + 7.2\%$, given a risk-free rate of $r_f=7.2\%$ over 2007–2011. We summarise our results in Table 2 and note the much higher risk-free rates r_f associated with the macroeconomic downturn period across both PL and ML portfolios. This phenomenon clearly reflects the market turbulence that prevailed during the downturn period, which caused bond yields to surge.

TABLE 2. Comparing various metrics across the PL and ML portfolios over two periods of time, having set the cost of capital rate at $c = \{6\%, 7\%, 8\%\}$. Metrics include the calculated risk-free rate r_f , the associated discount rate r_d , the resulting risk premium δ , the EC ratio EC/MCP, the average loss $\bar{L}$, and the sample standard deviation of losses $\hat{\sigma}$. Values in bold indicate the baseline scenario of $c=7\%$.

	c	r_f	r_d	δ	EC/MCP	$\bar{L}$	$\hat{\sigma}$
Personal loans							
Full period	6%	6.37%	13.90%	7.53%	1.441	0.747	0.293
	7%	6.37%	15.21%	8.84%	1.449	0.749	0.290
	8%	6.37%	16.52%	10.15%	1.455	0.751	0.288
Downturn period	6%	8.93%	15.86%	6.93%	1.424	0.758	0.275
	7%	8.93%	17.06%	8.13%	1.430	0.760	0.273
	8%	8.93%	18.27%	9.33%	1.436	0.762	0.270
Mortgage loans							
Full period	6%	6.88%	11.45%	4.57%	1.445	0.253	0.364
	7%	6.88%	12.28%	5.40%	1.468	0.256	0.365
	8%	6.88%	13.14%	6.26%	1.493	0.258	0.367
Downturn period	6%	8.49%	13.13%	4.64%	1.624	0.288	0.369
	7%	8.49%	13.99%	5.50%	1.653	0.290	0.370
	8%	8.49%	14.86%	6.37%	1.684	0.293	0.372

3. See the financial results of the Standard Bank Group, June 2012. Available at <https://tinyurl.com/364ayufz> (Accessed 16 May 2022)

Furthermore, we express the EC amount relative to the portfolio's MCP to enable comparison across portfolios and over different terms, a quantity we shall call the EC ratio. Evidently, this EC ratio is almost identical in both the PL and ML portfolios, despite having very different values for the average loss $\bar{L}$. The significantly longer workout period of the ML portfolio explains this fact, with an average time span (25.64 months) that is more than five times longer than that of the PL portfolio (4.67 months). Another factor is the greater standard deviation $\hat{\sigma}$ of L for the ML portfolio with its smaller losses, and vice versa for the PL portfolio. The greater average losses $\bar{L}$ of the PL portfolio also explains its greater discount rates r_d relative to that of the ML portfolio, which implies that the risk margin R affects r_d more stringently than the recoveries.

6.2 When comparing the downturn with the full period, the greater EC ratio (1.653 versus 1.469) of the ML portfolio is largely due to the portfolio's longer average workout period (33.59 versus 25.64 months) and its greater $\bar{L}$ value (29% versus 25.6%). However, this effect is offset by the greater r_f value, which results in risk premiums δ that are very similar (5.4% versus 5.5%) between the downturn and full period. The difference in r_d (15.21% versus 17.06%) is larger for the PL portfolio, which is almost entirely attributable to the greater difference in r_f values (8.93% versus 6.37%). Greater r_f values also explain why the PL portfolio's resulting δ value is lower during the downturn than the full period, which is otherwise curious. These results however corroborate that of Scheule & Jortzik (2020) in that lower δ values are associated with periods of greater r_f values, especially when applying the ME variant of the ROE method in determining r_d .

6.3 We highlight the following two interesting results in Table 2. Firstly, note the r_d rates across the downturn and full periods, which are respectively 13.99% and 12.28% for the ML portfolio, and 17.06% and 15.21% for the PL portfolio. The range of these r_d values aligns with those obtained by Scheule & Jortzik (2020), who analysed corporate loan recovery data from South Africa. Their estimated ROE-based r_d values range between 12% and 17% over the full period, where 17% coincided with the downturn period. Their average cost of equity of 14% also compares well with our chosen value of 14%. Moreover, their calculated equilibrium rates (or the ME variant of ROE within our context) range between 8% and 12.5%, and it peaks during mid-2008. The WACC-based r_d values follow the same pattern, but are approximately 50 basis points lower than the equilibrium rates over the same period. As for our second result, consider the cost of capital in Table 2 and compare its values across portfolios: a larger c value implies a greater r_d value. Moreover, it appears that c values have a much more pronounced effect on the r_d of the PL portfolio than that of the ML portfolio. This phenomenon is largely explained by the greater $\bar{L}$ values associated with the unsecured PL portfolio. In this case, the present value of the recoveries has a relatively lower weight in determining the r_d value than what would have been the case with the ML portfolio. In particular, a 1% increase in c results in an approximate increase of 1.2–1.3% in r_d for the PL portfolio, versus a 0.83–0.88% increase in r_d for the ML portfolio.

7. CONCLUSION

7.1 We have demonstrated how to infer a market-consistent discount rate r_d for LGD modelling purposes by calculating an MCP for a portfolio of defaulted debt. Determining such an MCP relies on the cost of capital that is required for holding defaulted debt in the first place, as inspired by the approach followed within the Solvency II regulatory regime. In our method, we calculated the EC for post-default loss risk by using the single factor model proposed by Tasche (2004). However, we anticipate that banks will likely use their own internal capital models for this purpose. Given an EC amount, we presented an algorithm for iteratively calculating the MCP as the difference between the cost of holding EC (or risk margin) and the discounted recoveries. In so doing, the risk premium δ is ultimately calculated within each iteration towards deriving a final r_d value.

7.2 We found that the main drivers of r_d include the following:

- the mean and variation of recovery cash flows;
- the average risk free rate r_f that applies over the reference period;
- the average term of the cash flows during the workout period; and
- the cost of capital rate.

As expected, greater volatility in the recovery cash flows is associated with greater risk, and hence, a higher r_d value. The same holds true for longer recovery terms. Similar to the findings of Scheule & Jortzik (2020), we found that a greater r_f value is, *ceteris paribus*, associated with a lower risk premium δ . This result renders the overall r_d value as surprisingly stable over multiple types of time periods, e.g., downturn versus full macroeconomic period. Furthermore, r_d is fairly sensitive to the cost of capital rate, and even more so when recovery rates are low. This sensitivity seems sensible when considering the cost of raising and servicing risk capital in guarding against variability in post-default recoveries. At the same time, consider that the cost of equity (in excess of the risk-free rate) should reflect a bank's overall riskiness and its cost of funding, which further justifies the strong link between r_d and the cost of capital.

7.3 As for future work, one might compare our resulting LGD values discounted by r_d to those discounted by the ever-popular average contract rate, particularly since our dataset prohibited such a comparison. It may also be useful to compare the two resulting LGD models that are built from these disparate LGD values, which are discounted differently. Ultimately, our proposed method satisfies the regulatory requirements of the Basel framework, in particular the self-styled Basel principle. This principle holds that r_d should account for both the time value of money, as well as include a risk premium for undiversifiable risk in the recovery cash flows. Our method can be easily applied throughout downturn and normal “through-the-cycle” macroeconomic conditions. We therefore deem our method appropriate in producing a suitable r_d value towards estimating the LGD component in both regulatory and economic capital modelling under the Basel framework.

DISCLOSURE OF INTEREST AND DECLARATION OF FUNDING

This work is not financially supported by any institution or grant, with no known conflicts of interest that may have influenced the outcome of this work. The authors would like to thank all anonymous referees and editors for their valuable contributions.

DATA AVAILABILITY STATEMENT

The data used in this study was provided by a large South African bank under a confidentiality agreement and cannot be shared publicly. However, to support transparency and replicability, we have included pseudocode in the manuscript that outlines the key computational steps.

REFERENCES

- Altman, EI & Pompeii, J (2003). Market size and investment performance of defaulted bonds and bank loans: 1987–2001. *Economic Notes*, **32**(2), 147–76
- Asarnow, E & Edwards, D (1995). Measuring loss on defaulted bank loans: A 24-year study. *Journal of Commercial Lending*, **77**(7), 11–23. http://www.defaultrisk.com/_pdf6j4/Measuring_Loss_Defaulted_Bnk_Lns-A_24-Yr_Stdy.pdf
- Baesens, B, Rösch, D & Scheule, H (2016). *Credit risk analytics: Measurement techniques, applications, and examples in SAS*. John Wiley & Sons
- BCBS (2003). The New Basel Capital Accord (tech. rep.). Bank for International Settlements. Switzerland (July). <http://www.bis.org/bcbs/irbriskweight.htm>
- BCBS (2004). An Explanatory Note on the Basel II IRB Risk Weight Functions (tech. rep.). Bank for International Settlements (January). Switzerland. <http://www.bis.org/bcbs/irbriskweight.htm>
- BCBS (2005). Guidance on paragraph 468 of the framework document. Bank for International Settlements. Switzerland. <https://www.bis.org/publ/bcbs115.htm>
- BCBS (2006). International convergence of capital measurement and capital standards: A revised framework, comprehensive version. Bank for International Settlements. Switzerland. <https://www.bis.org/publ/bcbs128.pdf>
- BCBS (2019). The Basel framework. Bank of International Settlements: Basel Committee on Banking Supervision (BCBS). Switzerland. https://www.bis.org/basel_framework/
- Boekel, P, van Delft, L, Hoshino, T, Ino, R, Reynolds, C & Verheugen, H (2009). Replicating portfolios. an introduction: Analysis and illustrations. Milliman Research Report. https://web.actuaries.ie/sites/default/files/erm-resources/replicating_portfolios_rr.pdf
- Botha, A (2021). A procedure for loss-optimising the timing of loan recovery under uncertainty (Doctoral dissertation, University of Pretoria). <https://doi.org/10.13140/RG.2.2.12015.30888/1>
- Brady, B, Chang, P, Miu, P, Ozdemir, B & Schwartz, D (2006). Discount rate for workout recoveries: An empirical study. SSRN Electronic Journal (Retrieved 10, 2020)
- Calabrese, R & Zenga, M (2010). Bank loan recovery rates: Measuring and nonparametric density estimation. *Journal of Banking and Finance*, **34**, 903–11

- Caselli, S, Gatti, S & Querci, F (2008). The sensitivity of the loss given default rate to systematic risk: New empirical evidence on bank loans. *Journal of Financial Services Research*, **34**, 1–34
- Dimson, E, Marsh, P & Staunton, M (2011). Equity premia around the world. Available at SSRN: <https://doi.org/10.2139/ssrn.1940165>
- Düllmann, K & Gehde-Trapp, M (2004). Systematic risk in recovery rates: An empirical analysis of US corporate credit exposures. Discussion Paper Series 2, No. 2004,02, Deutsche Bundesbank, Frankfurt a.M. https://www.econstor.eu/bitstream/10419/19729/1/200402dkp_b.pdf
- EBA (2017). GL/2017/16: Guidelines on PD estimation, LGD estimation and the treatment of defaulted exposures. European Banking Authority. France. <https://www.eba.europa.eu/sites/default/files/documents/10180/2033363/6b062012-45d6-4655-af04-801d26493ed0/Guidelines%20on%20PD%20and%20LGD%20estimation%20%28EBA-GL-2017-16%29.pdf>
- European Parliament. (2006). Directive 2006/48/EC of the European Parliament and of the Council of 14 June 2006 relating to the taking up and pursuit of the business of credit institutions. *Official Journal of the European Union*, **177**, 1–200
- European Parliament. (2009). Directive 2009/138/EC of the European Parliament and of the Council of 25 November 2009 on the taking-up and pursuit of the business of Insurance and Reinsurance (Solvency II). *Official Journal of the European Union L*, **335**, 1–155
- Fama, EF (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, **25**(2), 383–417. <https://doi.org/10.2307/2325486>
- Fama, EF & French, KR (2004). The capital asset pricing model: Theory and evidence. *Journal of economic perspectives*, **18**(3), 25–46. <https://doi.org/10.1257/0895330042162430>
- FASB (2016). Accounting standards update 2016–13, financial instruments – credit losses (Topic 326). Financial Accounting Standards Board. Norwalk, CT
- Federal Advisory Council and Board of Governors. (2015). FAC Meeting Minutes, May 8, 2015. <https://www.federalreserve.gov/aboutthefed/fac-20150512.pdf> (Accessed 16 January 2023)
- Ferguson, WL (2009). Solvency: Models, assessment and regulation. *Journal of Risk and Insurance*, **76**(4), 960
- Finlay, S (2010). *The management of consumer credit: Theory and practice* (Second ed.). Palgrave Macmillan
- Frye, J, Ashley, L, Bliss, R, Cahill, R, Calem, P, Foss, M, Gordy, M, Jones, D, Lemieux, C, Lesiak, M, Lobbes, C, McGrew, L, Mehta, P, Nelson, J & Waggoner, E (2000). Collateral damage: A source of systematic credit risk. *Risk*, **13**(4), 91–4
- Gibilaro, L & Mattarocci, G (2007). The selection of the discount rate in estimating loss given default. *Global Journal of Business Research*, **1**(1), 15–33
- Gordy, MB (2003). A risk-factor model foundation for ratings-based bank capital rules. *Journal of Financial Intermediation*, **12**(3), 199–232. [https://doi.org/10.1016/S1042-9573\(03\)00040-8](https://doi.org/10.1016/S1042-9573(03)00040-8)
- Gürtler, M & Hibbeln, M (2013). Improvements in loss given default forecasts for bank loans. *Journal of Banking & Finance*, **37**(7), 2354–66. <https://doi.org/10.1016/j.jbankfin.2013.01.031>
- Huang, C & Liang Xie, S (2012). Modeling loss given default (LGD) by finite mixture model. Proceedings of the Northeast SAS User Group 2012 Conference

- IASB (2011). International financial reporting standard (IFRS) 13: Fair value measurement. IFRS Foundation: International Accounting Standards Board. London. <https://www.ifrs.org/issued-standards/list-of-standards/ifrs-13-fair-value-measurement/>
- IASB (2014). International financial reporting standard (IFRS) 9: Financial instruments. IFRS Foundation: International Accounting Standards Board. London. <https://www.ifrs.org/content/dam/ifrs/publications/pdf-standards/english/2021/issued/part-a/ifrs-9-financial-instruments.pdf>
- Jacobs, M, Jr (2012). An empirical study of the returns on defaulted debt. *Applied Financial Economics*, **22**(7), 563–79
- Jankowitsch, R, Nagler, F & Subrahmanyam, MG (2014). The determinants of recovery rates in the US corporate bond market. *Journal of Financial Economics*, **114**(1), 155–77
- Larney, J, Grobler, GL & Allison, JS (2022). Introducing two parsimonious standard power mixture models for bimodal proportional data with application to loss given default. *Mathematics*, **10**(23), 4520
- Maclachlan, I (2005). Choosing the discount factor for estimating economic LGD. In Altman, EI, Resti, A & Sironi, A (Eds.), *Recovery risk: The next challenge in credit risk management*. Risk Books.
- Markowitz, H (1952). Portfolio selection. *The Journal of Finance*, **7**(1), 77–99. <https://doi.org/10.2307/2975974>
- Olbrys, J (2021). The global financial crisis 2007–2009: A Survey [SSRN Working Paper]
- Pelkiewicz, A, Ahmed, S, Fulcher, P, Johnson, K, Reynolds, S, Schneider, R & Scott, A (2020). A review of the risk margin – Solvency II and beyond. *British Actuarial Journal*, **25**, e1
- Phillips, R (2013). Optimizing prices for consumer credit. *Journal of Revenue & Pricing Management*, **12**(4), 360–77. <https://doi.org/10.1057/rpm.2013.9>
- PRA (2019). Cp21/19: Credit risk: Probability of default and loss given default. Bank of England, Prudential Regulation Authority. United Kingdom. <https://www.bankofengland.co.uk/-/media/boe/files/prudential-regulation/consultation-paper/2019/cp2119.pdf?la=en&hash=A241D325B94DEFFE3B5DDD7B87AEF9672ED9C439>
- PRA (2020). Ps11/20: Credit risk: Probability of default and loss given default. Bank of England, Prudential Regulation Authority. United Kingdom. <https://www.bankofengland.co.uk/-/media/boe/files/prudential-regulation/policy-statement/2020/ps1120.pdf?la=en&hash=99848DEA97DC50E27B71AEEB9F96F93C199BD9EE>
- Pykhtin, M (2003). Recovery rates: Unexpected recovery risk. *Risk*, **16**(8), 74–79
- SARB (2018). Long form credit report template (retail) [SARB, South Africa]. South African Reserve Bank. <https://www.resbank.co.za/content/dam/sarb/publications/prudential-authority/pa-long-form-review-templates/2018/8553/Retail-credit-long-form-template.docx>
- Scheule, HH & Jortzik, S (2020). Benchmarking loss given default discount rates. *Journal of Risk Model Validation*, **14**(3). <https://doi.org/10.21314/JRMV.2020.225>
- Schuermann, T (2004). What do we know about loss given default? Wharton Financial Institutions Center Working Paper No. 04-01. <http://dx.doi.org/10.2139/ssrn.525702>
- Sharpe, WF (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, **19**(3), 425–42

- Tasche, D (2004). The single risk factor approach to capital charges in case of correlated loss given default rates. <https://arxiv.org/abs/cond-mat/0402390>
- Thomas, LC (2009). *Consumer credit models: Pricing, profit and portfolios*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199232130.001.1>
- Van Gestel, T & Baesens, B (2009). *Credit risk management: Basic concepts*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199545117.001.0001>
- Vasicek, O (2002). The distribution of loan portfolio value. *Risk*, **15**(12), 160–2. <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.471.8181&rep=rep1&type=pdf>
- Weißbach, R & von Lieres und Wilkau, C (2010). Economic capital for nonperforming loans. *Financial Markets and Portfolio Management*, **24**(1), 67–85
- Witzany, J (2009). Unexpected recovery risk and LGD discount rate determination. *European Financial and Accounting Journal*, **4**(1), 61–84. <https://doi.org/10.18267/j.efaj.63>